
Exploring Deep Learning Training Paradigms and Model Architectures

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https://github.com/Ibrahim778/ATML_PA1

Abstract

This project explores the what and why behind the architectural decisions and training methods of several different deep learning models.

1 Convolutional Neural Networks (CNN) Training Comparisons

Table 1: Task 1: Clean baseline performance on the held-out test subset ($n = 240$).

Model	Accuracy	Macro-F1	Mean max. confidence
ResNet-50	0.972	0.972	0.947
ViT-B/16	0.970	0.970	0.965
CLIP ViT-B/32 (linear probe)	0.966	0.966	0.241
CLIP ViT-B/32 (zero-shot)	0.936	0.936	0.925

Table 2: Task 1: Effect of color interventions (grayscale, 90° hue rotation) on accuracy, prediction consistency vs. the clean prediction, and mean max. confidence.

Model	Intervention	Accuracy	Consistency vs. clean	Mean max. confidence
ResNet-50	Grayscale	0.946	0.952	0.943
	Hue rot. 90°	0.938	0.954	0.927
ViT-B/16	Grayscale	0.950	0.978	0.955
	Hue rot. 90°	0.934	0.958	0.938
CLIP ViT-B/32	Grayscale	0.938	0.940	0.225
	Hue rot. 90°	0.948	0.952	0.216

Table 3: Task 1: Shape-texture cue-conflict results.

Model	n_{shape}	$n_{texture}$	n_{other}	Shape bias (%)	Coverage (%)
ResNet-50	81	74	85	52.3	64.6
ViT-B/16	131	33	76	79.9	68.3
CLIP ViT-B/32	128	31	81	80.5	66.3

Table 4: Task 1: Accuracy and prediction consistency (vs. zero-shift) as a function of translation magnitude.

Model	0 px		8 px		16 px		32 px	
	Acc.	Cons.	Acc.	Cons.	Acc.	Cons.	Acc.	Cons.
ResNet-50	0.972	1.000	0.965	0.986	0.967	0.987	0.959	0.978
ViT-B/16	0.970	1.000	0.971	0.985	0.967	0.990	0.963	0.986
CLIP ViT-B/32	0.966	1.000	0.966	0.973	0.961	0.963	0.956	0.966

References

Table 5: Task 1: Accuracy and consistency vs. clean under patch shuffling.

Model	Accuracy	Consistency vs. clean
ResNet-50	0.888	0.898
ViT-B/16	0.914	0.936
CLIP ViT-B/32	0.832	0.818

Table 6: Task 1: Representation stability I_T (cosine similarity between clean and transformed features) by intervention.

Model	Grayscale	Translation (32 px)	Patch shuffle
ResNet-50	0.787	0.923	0.715
ViT-B/16	0.737	0.951	0.652
CLIP ViT-B/32	0.894	0.964	0.733